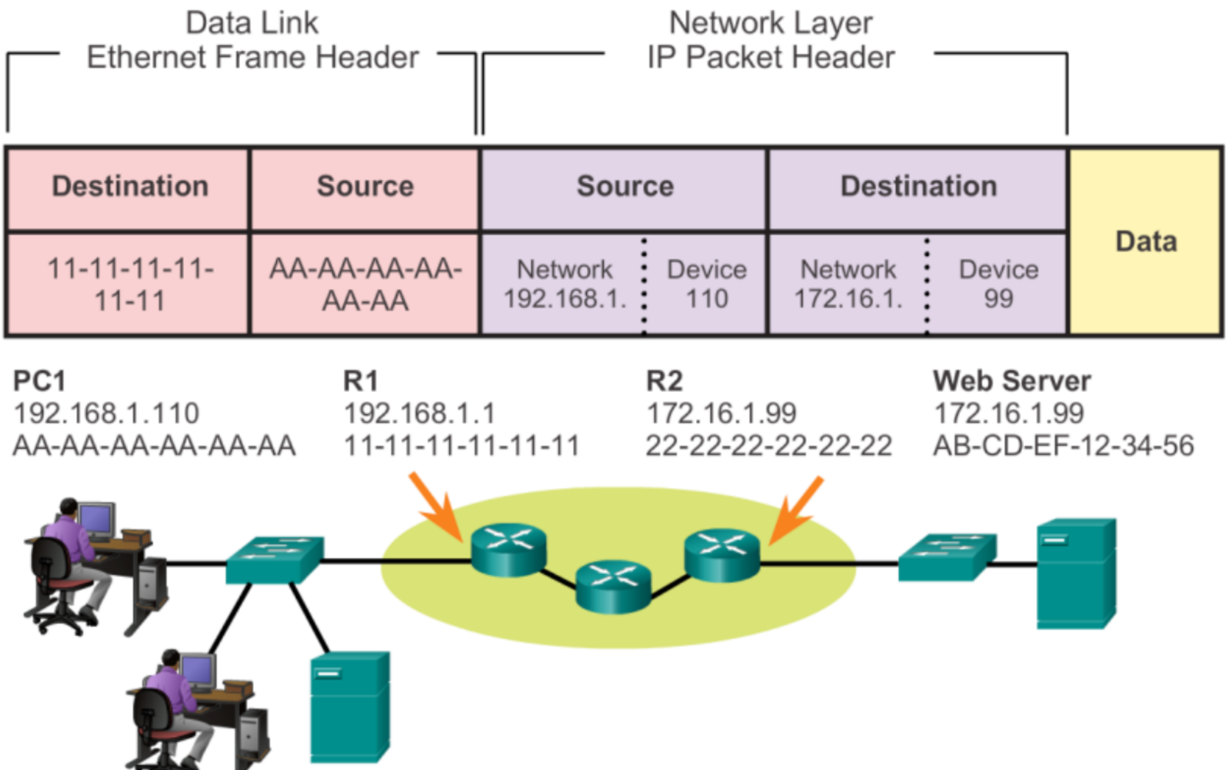
1 a) Consider a host (X) in a LAN wants to communicate another host (Y) in a remote LAN. Explain the ARP operation in such as a scenario with the necessary diagram. (10)

When two hosts (X and Y) are in different networks, the source host (X) cannot directly obtain the destination host's (Y) MAC address. Instead, it sends the packet to its default gateway (router), which forwards it to the destination network.



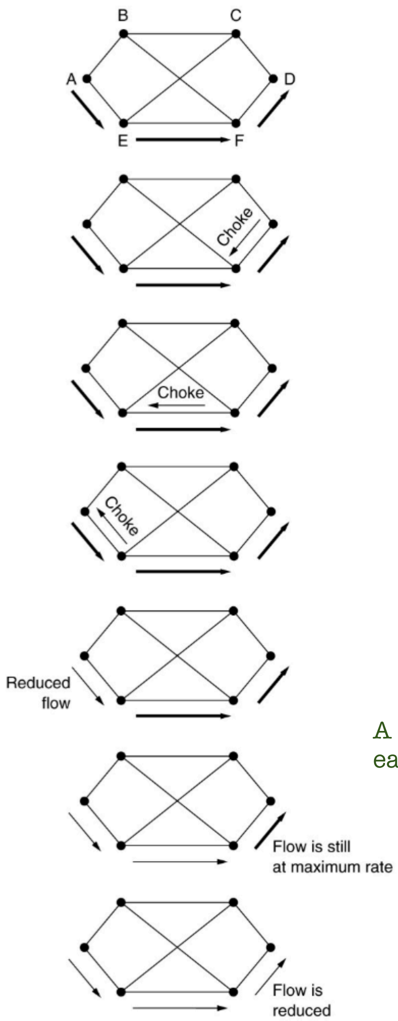
PC1 sends data to the Web Server but needs to forward it via R1 (default gateway).
PC1 finds R1's MAC via ARP and sends the packet with Src MAC: PC1 (AA-AA-AA-AA-AA-AA) and Dst MAC: R1 (11-11-11-11-11-11)
R1 forwards the packet to R2, updating the MAC addresses.
R2 checks routing and finds the Web Server's MAC via ARP.
R2 sends the packet to the Web Server with Src MAC: R2 (22-22-22-22-22-22) and Dst MAC: Web Server (AB-CD-EF-12-34-56)

b) Consider you are given the address space **172.16.10.0/23.** Now you are to subnet the address according to the host requirements for different departments that are given in the following table. Then fill up the empty columns with values resulting from your calculation. Use the smallest subnet sizes that will accommodate the relevant number of hosts and do not leave any gap (at the start of given address space or between the subnets). (20)

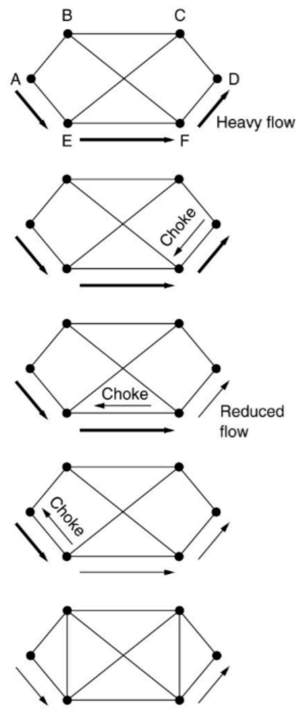
Department	No. of hosts required	Network address	Broadcast Address	Subnet Mask
Sales	100	172.16.10.0	172.16.10.127	255.255.255.128
Service	90	172.16.10.128	172.16.10.255	255.255.255.128
Accounts	30	172.16.11.0	172.16.11.31	255.255.255.224
Sales	12	172.16.11.32	172.16.11.47	255.255.255.240
HR	10	172.16.11.48	172.16.11.63	255.255.255.240

2a) "Sending hop-by-hop choke packets performs better than sending the choke packet to the sender"- Do you agree with the statement? Justify your opinion with necessary figure(s). (10)

Yes, sending hop-by-hop choke packets generally performs better than sending the choke packet directly to the sender.
In hop-by-hop choke packet mechanism, congestion is controlled at each intermediate router, preventing further congestion from building up along the path.
If the choke packet is sent directly to the sender, congestion continues until the sender receives and reacts to it, which takes more time.
If a router detects congestion, it can slow down the transmission from its previous hop immediately rather than waiting for a distant sender to react.
This reduces packet drops and improves overall network performance.



A choke packet that affects only the source.



A choke packet that affects each hop it passes through.

b) What are the different tables required for OSPF routing? How can a router build those tables during the OSPF operation? Which table contributes to provide full network structure to each OSPF router? (10)

Database	Table	Description
Adjacency Database	Neighbor Table	• List of all neighbor routers to which a router has established bidirectional communication. • This table is unique for each router. • Can be viewed using the **`show ip ospf neighbor`** command.
Link-state Database (LSDB)	Topology Table	• Lists information about all other routers in the network. • The database shows the network topology. • All routers within an area have identical LSDB. • Can be viewed using the **`show ip ospf database`** command.
Forwarding Database	Routing Table	• List of routes generated when an algorithm is run on the link-state database. • Each router's routing table is unique and contains information on how and where to send packets to other routers. • Can be viewed using the **`show ip route`** command.

The Topology Table provides the full network structure.

c) What can be a possible solution to Count-to-infinity problem? Explain briefly. (10)

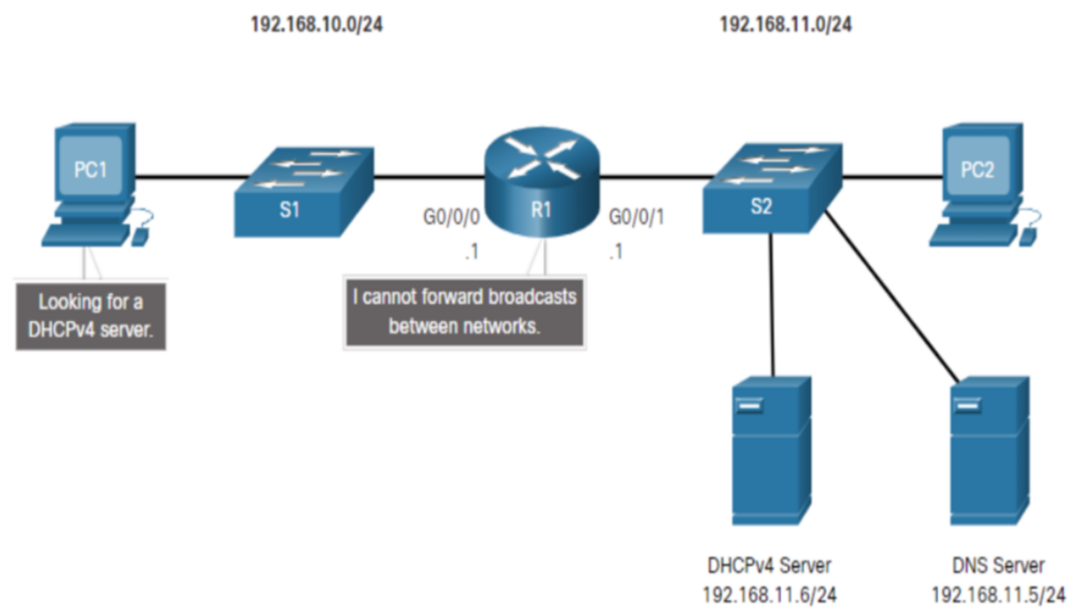
Counting-to-infinity is just another name for a routing loop. In distance vector routing, routing loops usually occur when an interface goes down. It can also occur when two routers send updates to each other at the same time.

Loop Breaking Heuristics:

1. Set infinity to a limited number, e.g. 16.
2. Split horizon (A router does not advertise a route back to the neighbor from which it learned it)
3. Split horizon with poison reverse

3a) What is the purpose of DHCP relay agent? Show its operation using a topology diagram. (10)

In a complex hierarchical network, enterprise servers are usually located centrally. These servers may provide DHCP, DNS, TFTP, and FTP services for the network. Network clients are not typically on the same subnet as those servers. In order to locate the servers and receive services, clients often use broadcast messages.

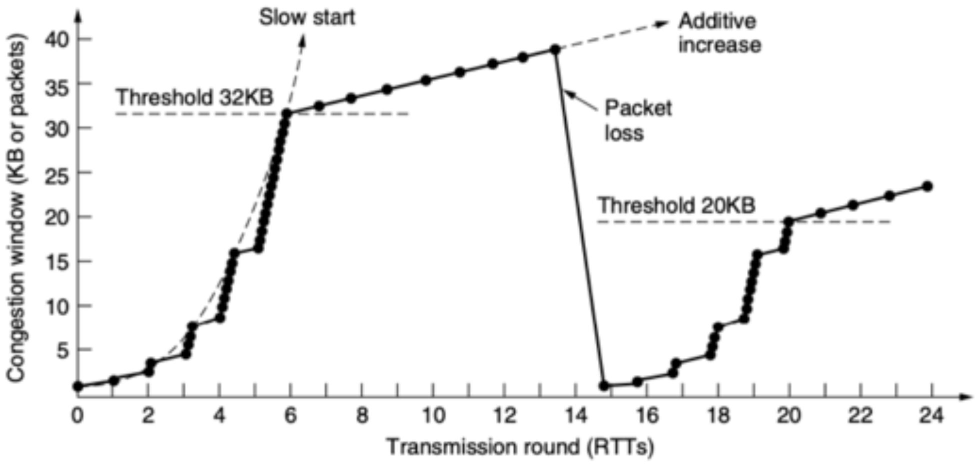


In the figure, PC1 is attempting to acquire an IPv4 address from a DHCPv4 server using a broadcast message. In this scenario, R1 is not configured as a DHCPv4 server and does not forward the broadcast. Because the DHCPv4 server is located on a different network, PC1 cannot receive an IP address using DHCP. R1 must be configured to relay DHCPv4 messages to the DHCPv4 server.

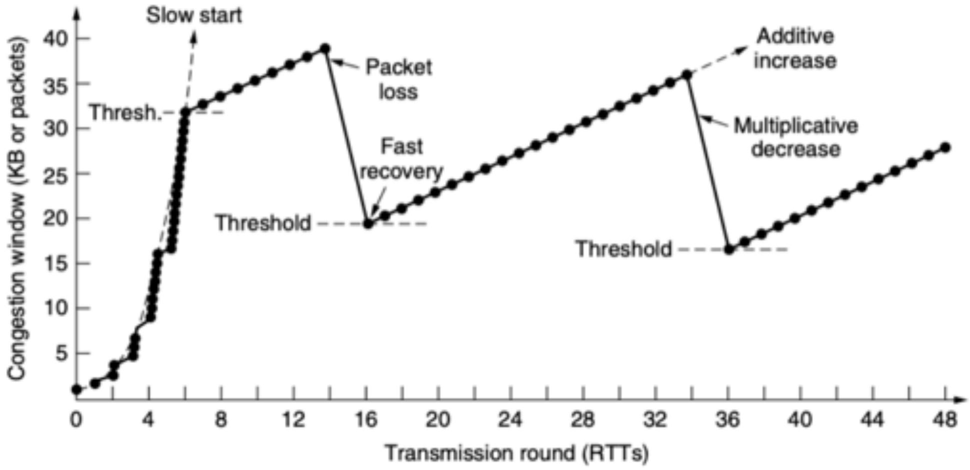
When R1 has been configured as a DHCPv4 relay agent, it accepts broadcast requests for the DHCPv4 service and then forwards those requests as a unicast to the IPv4 address 192.168.11.6.

b) Compare TCP Tahoe with TCP Reno congestion control algorithm with necessary figure(s). (10)

TCP Tahoe: If 3 duplicate ACKs are received, CWND is reset to 1, and Slow Start begins.



TCP Reno: If 3 duplicate ACKs are received, CWND is halved, and Fast Recovery is used to maintain throughput.



c) In a TCP session, the receiver sends ACK=1 with a sequence number 8192 and Window Size 0. What does it imply to the sender? (10)

ACK=1 and Sequence Number 8192:
The receiver has successfully received all data up to sequence number 8191.
The receiver is now expecting data starting from sequence number 8192.

Window Size = 0:
The receiver's buffer is full, and it cannot accept any more data at the moment.
The sender must stop sending data until the receiver advertises a non-zero window size.

4a) What is non-authoritative DNS resolution? When can it happen? Do you trust such resolution? (10)

Reduction in DNS lookup can be reduced at a greater scale if the mapping is stored locally. When a server asks for a mapping from another server and receives response, it stores the information in its cache memory before sending it to client to satisfy other clients afterwards without further query The response coming from the cache is marked as **Unauthoritative/non-authoritative**.

Non-authoritative responses are generally reliable if the DNS resolver is trustworthy (e.g., Google DNS, Cloudflare). However, cached responses may be outdated if the domain's DNS records have changed recently (before the TTL expires).

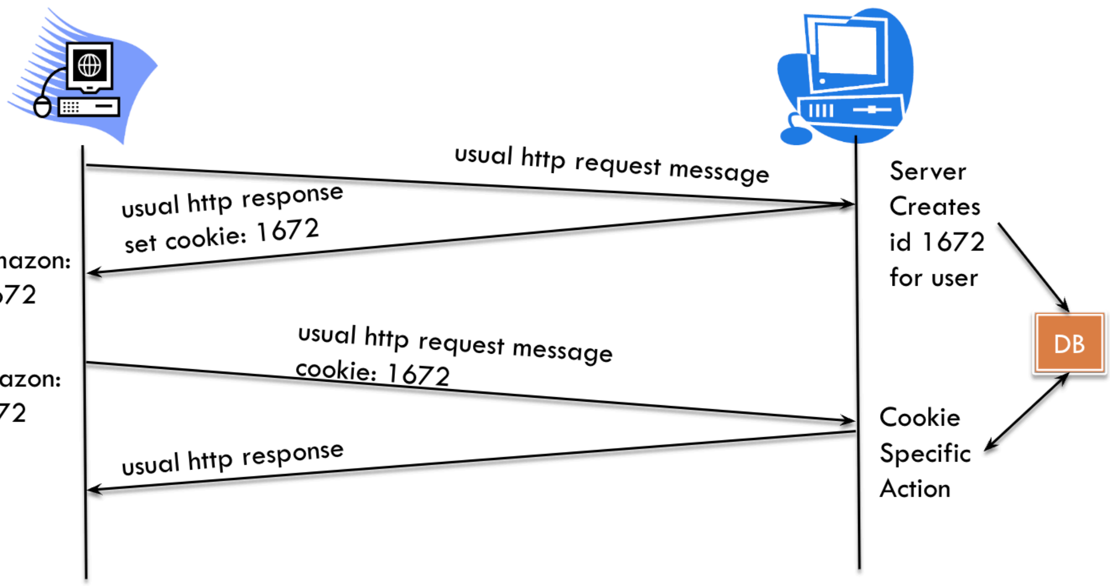
b) What is the purpose of cookies in HTTP communication? What are its benefits and drawbacks? (10)

Cookies are small pieces of data stored on a user's device by a web server during HTTP communication.

Purpose: Used to maintain state in HTTP (which is otherwise stateless) for session management, personalization, and tracking.

Benefits: Authorization, shopping carts recommendation, user session state.

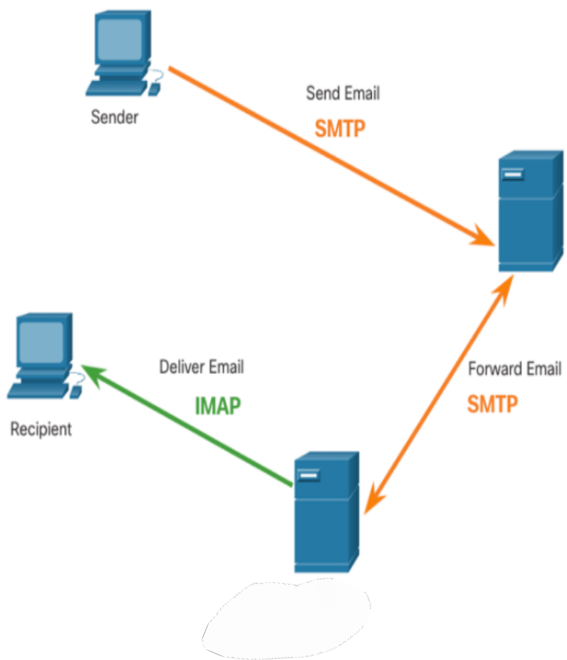
Drawbacks: Websites learns a lot about user.



c) Explain the roles of SMTP and IMAP protocols in email communication using necessary figure. (10)

Simple Mail Transfer Protocol (SMTP):

It is a simple ASCHII protocol. After establishing a TCP connection to port 25, the sending machine (client) waits for the receiving machine (server). The server starts by sending a line of text giving its identity and telling whether it is prepared to receive mail. If the server is not ready, the client releases the connection and tries again. If the server is willing to accept e-mail, the client announces whom the e-mail is coming from and whom it is going to. If such recipients exists at the destination, the server gives the client to go-ahead to send the message. Then the client sends the message and server acknowledges it.



Internet Message Access Protocol (IMAP)

It assumes all e-mail will remain on the server indefinitely in multipart mailboxes. It provides mechanism for reading messages or even part of messages. It also provides mechanism of creating, destroying and manipulating multiple mailboxes on the server. When a user connects to an IMAP server, copies of the messages are downloaded to the client application. The original messages are kept on the server until manually deleted. When a user decides to delete a message, the server synchronizes that action and deletes the message from the server.